



Thiopeptides encoding biosynthetic gene clusters mined from bacterial genomes

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Currently, we are at the threshold of the ‘post-antibiotic era’ due to the global emergence of antimicrobial resistance (AMR), and hence there is a dire need to discover new antibiotics. Ribosomally synthesized and post-translationally modified peptides (RiPPs) are a diverse class of natural products (NPs), some of which are under clinical trials for their antimicrobial potential. Thiopeptides are structurally one of the most complex classes of RiPPs due to numerous post-translational modifications (PTMs), with [4+2] cycloaddition being the core PTM and are active against several gram-positive pathogens. Genome mining coupled with experimental work can harness the unexplored ‘cryptic’ gene clusters while minimizing the rate of the rediscovery of known metabolites and expand the molecular diversity of NPs with medicinal potential. Employing the genome mining approach using a series of freely available bioinformatics tools, we have identified eight novel putative thiopeptide encoding biosynthetic gene clusters (BGCs) from different bacterial genomes, most of which belong to the class Actinobacteria. Our results provide confidence in the newly identified BGCs, to proceed with wet-bench experiments and discover novel thiopeptide(s).

Keywords. Genome mining; RiPP; thiopeptide; biosynthetic gene cluster; [4+2] cycloaddition enzyme

1. Introduction

With recent advances in bioinformatics and genome mining, the biosynthetic potential of microbes can be explored *in silico*, and new ribosomal natural products (NP) can easily be discovered with minimal wet-lab experiments. Secondary metabolites from microorganisms that have been providing bacteria as a safeguard are a propitious source for determining bioactive NPs with therapeutic potential for future drug development (Zhong *et al.* 2020). Bacterial infections are increasing the mortality rate day by day due to a phenomenon called antimicrobial resistance and if this phenomenon persists, the number can rise to 10 million deaths per year by 2050 (Schneider *et al.* 2018). Infections caused by antibiotic-resistant pathogens like VRE (vancomycin-resistant *Enterococci*), MRSA (methicillin-

resistant *Staphylococcus aureus*), and MDR-TB (multidrug-resistant *Mycobacterium tuberculosis*) are rarely curable as of now; therefore, there is a need to develop new antibiotics.

The diversity in structures and functions of ribosomally synthesized and post-translationally modified peptides (RiPPs) attract considerable attention from both industrial as well as academic institutions. RiPP biosynthetic gene clusters (BGCs) generally code for a short precursor peptide divided into an ‘N-terminal leader’ and a ‘C-terminal core peptide’ and the post-translational modification (PTM) enzymes. RiPPs include different classes like lanthipeptides, thiopeptides, lassopeptides, microviridins, sactipeptides, LAP (linear thiazole/oxazole-containing peptides), botromycins, and many more that can be obtained from different taxa of bacteria, fungi, archaea, and plants

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(Zhong *et al.* 2020). One of the medicinally promising classes of RiPP is thiopeptides (also known as thiazolyl peptides) which suppress the growth of gram-positive bacteria by blocking their protein translation (Schwalen *et al.* 2018). Thiopeptides have been used in animal feed for a long time because of their low toxicity and they have the potential to be a ‘lead compound’ for drug development. The core modifications of thiopeptides comprise five-membered sulfur (or oxygen) heterocycles, dehydroserines (or dehydrothreonines), and six-membered azacycles (N-heterocycles) (Zheng *et al.* 2017). Many studies have explained that a well-arranged succession of PTMs lead to the complex structural hallmarks of thiopeptides.

Thiopeptide biosynthesis (figure 1) begins with the ribosomal synthesis of a 50–60 amino acids long precursor peptide. The core region constitutes mature thiopeptide, and the leader region which is generally 30–40 amino acids long, serves as a recognition scaffold for PTM enzymes and gets cleaved off proteolytically at a specific position to give rise to an active peptide (Vinogradov and Suga 2020). The PTM enzymes for the thiopeptide biosynthesis include cyclodehydratase (YcaO), a dehydrogenase, a Dehydratase, and a [4+2] cycloaddition enzyme (Montalbán-López *et al.* 2020). First, cyclodehydration of cysteine and a few serine/threonine residues, is carried

out by the Ocin-ThiF-dependent YcaO to form azoline heterocycles, which is often paired with subsequent dehydrogenation to the corresponding azole. Then, split LanB-like dehydratase acts on unmodified serine and threonine residues in a tRNA-dependent fashion and gives rise to dehydroalanine (Dha) and dehydrobutyrine (Dhb), respectively. Lastly, the [4+2] cycloaddition/macrocyclization enzyme takes over and a formal [4+2] cycloaddition occurs between two Dha residues to form a six-membered central N-heterocycle, which is common in all thiopeptides. Generally, this heterocycle is present in the pyridine state, but other states like hydroxypyridine, piperidine, and dehydropiperidine are also known (Schwalen *et al.* 2018).

The availability of ever-expanding genome data and refined bioinformatic tools have brought a revolutionary change in the discovery of novel RiPPs, even if their native producer is difficult to culture (Hetrick and van der Donk 2017). We discovered novel thiopeptides by employing the classical approach to use one of the core biosynthetic enzymes as a driving sequence to search homologs, by using a series of freely available bioinformatics software such as the antiSMASH, an integrated platform of general propose for BGCs prediction and analysis (Blin *et al.* 2019), RiPPMiner (Agrawal *et al.* 2017), and DeepRiPP (Merwin *et al.* 2020) for recognizing RiPP precursor peptides and prediction of their cleavage sites.

As done previously, where a sequence of class I lanthipeptide dehydratase from goadsporin BGC was used as a query to identify novel thiopeptide gene clusters (Hayashi *et al.* 2014), we adopted the same strategy and used split LanB-like dehydratase (TsrD) as a query sequence, to find novel thiopeptide gene clusters. It resulted in the identification of four BGCs from four different bacterial strains, selected randomly from the first 100 hits. But in a recent review (Montalbán-López *et al.* 2020), [4+2] cycloaddition enzyme has been reported as a class-defining enzyme for thiopeptide biosynthesis. We realized that two of our four BGCs were found not to code for [4+2] cycloaddition enzyme, so we screened more bacterial genomes randomly, from a non-redundant list of PSI-BLAST, but using the key enzyme [4+2] cycloaddition (TsrE) as a query and found four more BGCs from different bacterial strains.

Hence, we identified eight novel RiPP BGCs and twenty precursor peptides. Out of eight BGCs, only six have the genes to code for all the enzymes required for thiopeptide biosynthesis and thus should be capable of forming thiopeptide(s).

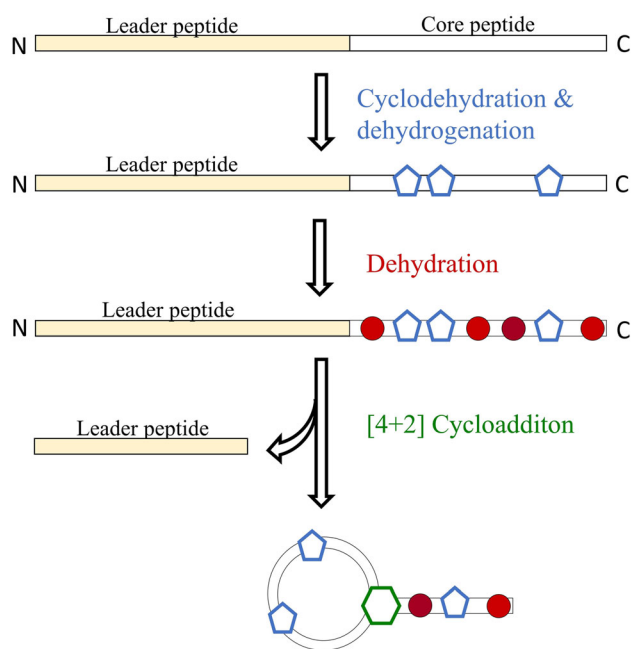


Figure 1. The schematic representation of common PTMs involved in thiopeptide biosynthesis.

2. Methods

2.1 Mining the homologs

TsrD (ACN52294.1) and TsrE (ACN52295.1) from *S. laurentii* were used as a query for PSI-BLAST (Position-Specific Iterative BLAST; <https://blast.ncbi.nlm.nih.gov/Blast.cgi>) to pull out the homologs with their respective e-values (supplementary tables 1 and 2). The novelty of the selected homologs was checked using MIBiG (Minimum Information about a Biosynthetic Gene cluster; Kautsar *et al.* 2019; <https://mibig.secondarymetabolites.org>) and NCBI (National Center for Biotechnology Information) genome database.

2.2 RiPP BGCs identification and prediction

The whole genome of the selected bacteria was subjected to the antiSMASH 5.0 (antibiotics and Secondary Metabolites Analysis Shell; <https://antismash.secondarymetabolites.org/#!/start>) for analyzing the presence of RiPP BGCs using default parameters. The amino acid or nucleotide sequence of each gene obtained from the thiopeptide coding region from the antiSMASH was subjected to CDD (conserved domain database; <https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) to determine their superfamilies which can further be used to interpret the protein functions. The genes smaller than 250 nucleotides were subjected to RiPPMiner to determine any similarity with known thiopeptides. The final BGCs for the thiopeptide coding genome were drawn by comparing their superfamilies with Thiostrepton A biosynthetic gene cluster (figure 2). Conserved amino acids in putative precursors were identified by performing ClustalW sequence alignment using MEGA-X (Kumar *et al.* 2018).

2.3 Confirmatory analysis

Thiopeptide encoding nucleotide region predicted by the antiSMASH was submitted in ORFfinder (Open Reading Frame Finder; <https://www.ncbi.nlm.nih.gov/orffinder>). The ORFs for query homologs were hunted in the genome sequence, and for precursor peptide, the ORFs ranging from 25 to 90 amino acids on the same strand were also checked. DeepRiPP (<http://deepripp.magarveylab.ca>) was used for identifying RiPP classes with class prediction probability and cleaved sequence.

3. Results

To dig out novel thiopeptide encoding clusters, a bioinformatics based approach was implemented. Using Split LanB-like dehydratase (TsrD) and [4+2] cycloaddition enzyme (TsrE) as the driver sequences, novel putative thiopeptide biosynthetic clusters from the selected bacterial genomes were identified. In our study, the hits obtained after PSI-BLAST were found among the representatives of Actinobacteria, Proteobacteria, Chloroflexi, Bacteroidetes, Cyanobacteria, and Firmicutes. However, more than 75% of strains bearing homologs belonged to the Phylum Actinobacteria (figure 3). The genes for modification enzymes and precursor peptides were identified in eight bacterial strains.

3.1 Identified BGCs using TsrD homologs as lead

3.1.1 *Planomonospora sphaerica* NRRL 18923: *P. sphaerica*, a member of the ‘rare’ genus of family Actinomycetes first isolated from Indian soil having 72.73% of GC content is a potential antibiotic producer (Mertz 1994). PSI-BLAST detected a TsrD homolog showing 69.59% sequence identity and e-value:

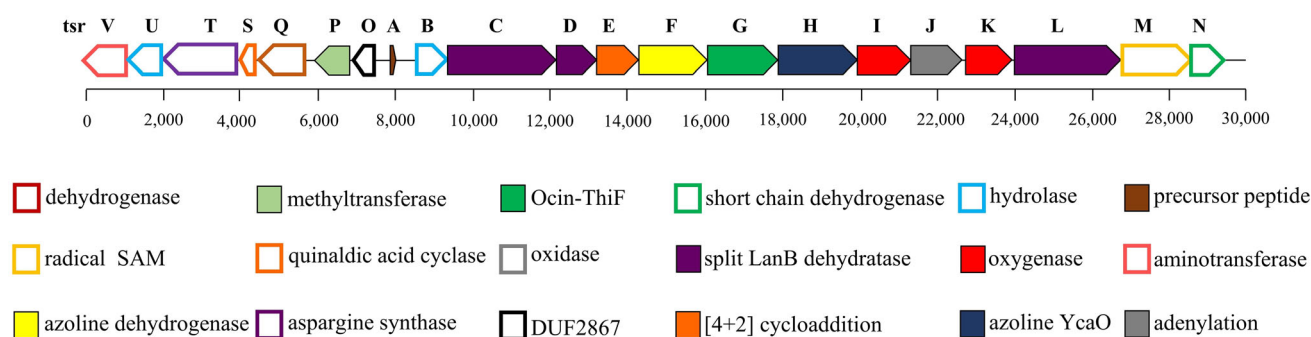


Figure 2. Thiostrepton A biosynthetic gene cluster that was used for comparison and determining the functions of the genes mined in this study.

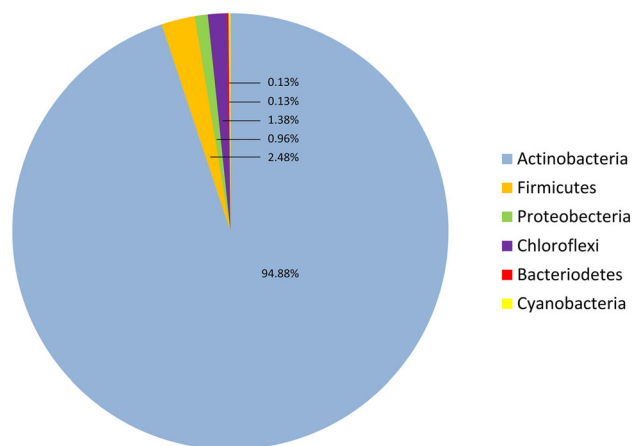


Figure 3. Share of phyla in the non-redundant list of 723 homologs (for both TsrD and TsrE combined together) obtained through PSI-BLAST. Percentage of cultures in each phylum is as follows: Actinobacteria: 94.88%; Firmicutes: 2.48%; Proteobacteria: 0.96%; Chloroflexi: 1.38%; Bacterioidetes: 0.13%; and Cyanobacteria: 0.13%.

4.00E–153 (supplementary table 1) originally named as ‘lantibiotic biosynthesis protein’ (GAT69287.1) in NCBI protein database. When its genome (BDCX01000013.1) (Dohra *et al.* 2016) was subjected to the antiSMASH, it showed a thiopeptide BGC (Cleavage pHMM score: 22.10, RODEO score: 32) having different genes (figure 4). Superfamilies of genes were predicted by CDD and functions of proteins were speculated based on comparison with BGC of Thiostrepton A. The thiopeptide coding region from the antiSMASH consists of cyclodehydration, macrocyclization, several other enzymes, and a single putative precursor peptide (ctg1_22) which was predicted as thiopeptide by RiPPMiner (77.36% identical to Thiostrepton A with e-value: 2E–18 and 50.00% identical to Nosiheptide with e-value: 5E–04). ORF finder exhibited the ORFs for both, the TsrD homolog and the precursor peptide on the negative-strand of the genome sequence. DeepRiPP was used to confirm the class of the precursor, which predicted it as ‘thiopeptide’ with 100% class prediction probability.

ClustalW alignment of the identified putative precursor peptide with Thiostrepton A and Nosiheptide indicated the presence of conserved amino acids especially Cys, Ser, Thr which can be possible targets for cyclodehydration, dehydration, or macrocyclization (figure 6A). As it was known that in Thiostrepton A, leader peptide cleavage occurs after M (Bennallack and Griffiths 2017), the possibility of cleavage at the same very site in this putative precursor peptide was confirmed by DeepRiPP. However, alignment with Nosiheptide suggests that cleavage can also occur after SA

of the conserved motif SAS (Bennallack and Griffiths 2017). Therefore, the actual cleavage for the putative precursor peptide can only be determined by performing *in vitro* experiments. These analyses suggested that the cluster is a putative thiopeptide cluster.

3.1.2 *Micromonospora carbonacea* NRRL 2972: *M. carbonacea* is a producer of two oligosaccharide antibiotics: Everninomicin B and D. It forms a branched mycelium and exists as a saprophyte in soil and water (Sanders and Sanders 1974). The genome analysis of *M. carbonacea* (NZ_FMC01000006.1) from NCBI showed a homolog of TsrD originally annotated as ‘hypothetical protein’ (WP_176734888.1) showing 59.50% sequence identity and e-value: 6.00E–121 (supplementary table 1) with TsrD in PSI-BLAST. The antiSMASH analysis (Cleavage pHMM score: 20.50; RODEO score: 27) showed a BGC (figure 4) with genes coding for thiopeptide modification enzymes and a putative precursor peptide (ctg1_401) that showed 77.78% sequence identity and e-value: 3E–15 with Thiostrepton A as indicated by RiPPMiner. Many other transcriptional regulators (TR) were also present upstream and downstream to the TsrD homolog. Individual ORFs were found for both, the TsrD homolog as well as for the putative precursor peptide on the negative-strand of the genome sequence. The class of putative precursor was confirmed as ‘thiopeptide’ by using DeepRiPP as it showed 100% class prediction probability.

Pairwise sequence alignment of the identified putative precursor peptide revealed high sequence similarity with Thiostrepton A, which suggests the presence of a similar leader peptide cleavage site (figure 6), but this is contradictory to the results shown by DeepRiPP, where the cleaved sequence starts with Serine residue of SAS motif. Hence, the leader peptide cleavage site of the identified putative precursor peptide has to be confirmed, for it to be established as a thiopeptide.

3.1.3 *Thermobifida halotolerans* YIM 90462: This actinomycete strain is an aerobic gram-positive bacteria that is thermotolerant and halotolerant which was first discovered from a saline soil sample from Yunnan, China (Yang *et al.* 2008). PSI-BLAST (supplementary table 1) results from non-redundant database showed a TsrD homolog of 341 amino acids (30.63% sequence identity; e-value: 4.00E–31), annotated as ‘lantibiotic biosynthesis protein’ in the protein database (WP_084012656.1). The antiSMASH result of its genome (NZ_LIZN01000011.1) revealed a cluster (figure 4) with the result ‘No core peptide found’

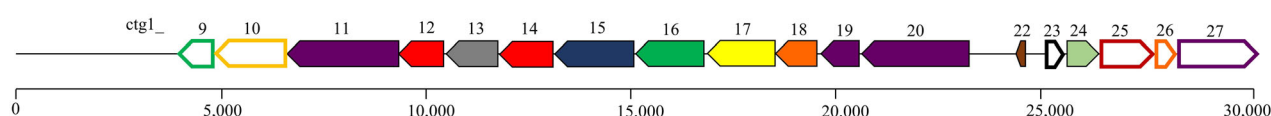
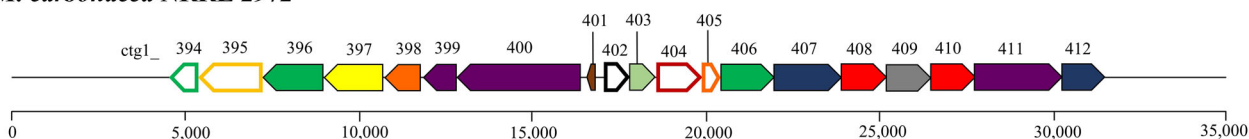
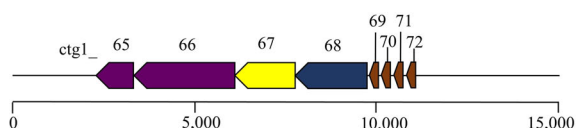
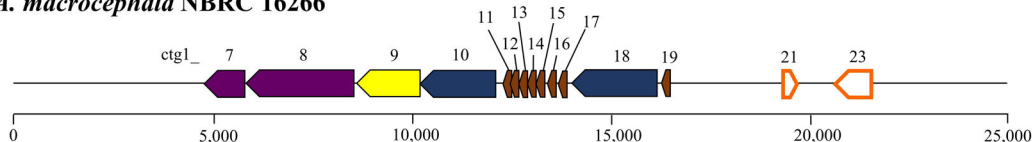
P. sphaerica* NRRL18923**M. carbonacea* NRRL 2972*****T. halotolerans* YIM 90462*****A. macrocephala* NBRC 16266**

Figure 4. Identified RiPP BGCs from different bacterial genomes using TsrD as a query. *P. sphaerica* NRRL18923 and *M. carbonacea* NRRL2972 are capable of forming thiopeptide, whereas *T. halotolerans* YIM90462 and *A. macrocephala* NBRC 16266 do not seem to be capable of thiopeptide biosynthesis as their BGCs lack the class-defining [4+2] cycloaddition enzyme. Gene clusters of the detected BGCs are in comparison to the known thiopeptide, Thiosrepton A (generated by the antiSMASH).

(Cleavage pHMM score and RODEO score: not mentioned). After subjecting each gene to CDD and comparing their superfamilies with Thiostrepton BGC manually, the proteins were identified as: azoline YcaO, azoline dehydrogenase, split LanB-like dehydratase except [4+2] cycloaddition enzyme. Even though the antiSMASH couldn't predict the precursor peptide encoding gene, the genes with less than 250 nucleotides (ctg1_69, 70, 71, 72) showed 40%, 40%, 24%, and 23.64% sequence identity respectively with Thiocillin GE37468 (e-values: 0.25, 0.25, 0.13, and 0.023 respectively) in RiPPMiner. So, we inferred that these genes may code for the putative thiopeptide precursors. The TsrD homolog and peptides were further confirmed using ORFfinder and we could find only the first three peptides on the negative-strand. But four

genes encoding peptides were identified, thus to find the fourth one, we changed parameters of ORFfinder to 'ATG and alternative initiation codons' and could locate the fourth one as well. Finally, to confirm their class and cleaved sequences, DeepRiPP was used, which showed peptides homology to 'thiopeptide' with 100% class prediction probability for all.

The absence of gene encoding [4+2] cycloaddition enzyme indicates that this particular BGC might not code for a thiopeptide, as [4+2] cycloaddition is a class-defining enzyme according to a recently published review article (Montalbán-López *et al.* 2020). We inferred that this BGC might code for another class of RiPP, i.e. LAP (e.g. Goadsporin) since it requires similar PTM enzymes as azoline YcaO, dehydrogenase, and split dehydratase (Montalbán-López *et al.*

2020). The above software (DeepRiPP) predicted the precursors as thiopeptide, and hence we cannot rely on this and experimental affirmation only will help in concluding whether the identified cluster produces a thiopeptide or a LAP.

3.1.4 *Acrocarpospora macrocephala* NBRC 16266: *A. macrocephala*, a gram-positive spore-bearing, and a strictly aerobic strain is a soil isolate from Saitama, Japan (Tamura *et al.* 2000). Genomic annotation of *A. macrocephala* showed ‘lantibiotic biosynthesis protein’ (GES11969.1, 33.23% identical to TsrD; e-value: $1.00\text{E}-41$) as the homolog in PSI-BLAST (supplementary table 1). The genes of thiopeptide encoding BGC, as predicted by the antiSMASH, were subjected to CDD analysis which showed presence for all PTM enzymes except for [4+2] cycloaddition enzyme. Eight genes (*ctg1_11*, 12, 13, 14, 15, 16, 17, 19) smaller than 250 bp were also present in the BGC (figure 4). Out of the eight, only *ctg1_11*, 12, 13, 15, 16, and 17 were showing 47.37%, 47.37%, 40.00%, 42.86%, 27.59% sequence identity to Thiocillin (e-value: 0.12, 0.30, 0.06, 1.2, and 3.1) respectively and 33.33% sequence identity to Thiomuracin A (e-value: 0.46) in RiPPMiner; so, these may code for putative thiopeptide precursors. For confirmation, we submitted the thiopeptide encoding region of the genome in ORFfinder and we found split LanB-like dehydratase homolog and ORF for all the genes on the negative-strand which were already identified except *ctg1_14*. For locating the ORF for *ctg1_14*, we picked only the thiopeptide coding region, 12278 to 13291 base pairs from the whole genome and fed it in ORFfinder, and found it. DeepRiPP was also used and *ctg1_12*, 13, 14, 15, 16, 17 and 19 were depicted as ‘thiopeptide’ with 97.70%, 63.06%, 97.10%, 99.22%, 99.89%, 99.96%, and 100% class prediction probability respectively.

Our analysis indicated that the above BGC can be a producer of LAP as the class-defining [4+2] cycloaddition enzyme for thiopeptide is missing here too. Wet-bench experiments are needed for determining the actual product of this BGC. Henceforth, a further search was done using TsrE as the sequence query to specifically mine the thiopeptide encoding BGCs.

3.2 Identified BGCs using TsrE homologs as lead

3.2.1 *Thermogemmatisspora onikobensis* NBRC 111776: *T. onikobensis*, a member of class Ktedonobacteria, is a thermophilic gram-positive and sporulating bacteria (Yabe *et al.* 2011). The genome

(Komaki *et al.* 2016) analysis of *T. onikobensis* (NZ_BDGT01000006.1) from NCBI showed a TsrE homolog, originally annotated as ‘hypothetical protein’ (WP_069802514.1) showing 24.6% sequence identity and e-value: $6.00\text{E}-11$ (supplementary table 2) in PSI-BLAST. The antiSMASH analysis displayed a BGC (figure 5) but with the result ‘No core peptides found’ (Cleavage pHMM score and RODEO score: not mentioned). Subjecting each of the genes of the identified cluster to CDD analysis helped in the identification of modification enzymes for thiopeptide formation. Even though the antiSMASH could not predict the precursor peptide encoding genes but the presence of thiopeptide modification enzymes compelled us to look for the presence of genes for precursor peptide(s) in the near vicinity. Therefore, genes with less than 250 nucleotides (*ctg1_120*, 121, 122, 123) were individually analyzed RiPPMiner to check their identity with any known thiopeptide. Three genes (*ctg1_120*, 121, 122) showed 39.98%, 39.29%, and 42.31% sequence identity and e-value: $3\text{E}-06$, $1\text{E}-06$, and $2\text{E}-07$ respectively with Thiomuracin A and the gene (*ctg1_123*) showed 61.54% sequence identity and e-value: 6.4 with Streptin2 (lanthipeptide A), clearly concluding (*ctg1_120*, 121, and 122) that these genes may code for thiopeptide precursors except one gene (*ctg1_123*). On the negative-strand of the genome sequence, the presence of the [4+2] cycloaddition enzyme homolog and all three putative peptides, were confirmed by ORFfinder. DeepRiPP predicted the three precursors as ‘thiopeptide’ with class prediction probability: 100% for all three peptides.

Multiple sequence alignment (figure 6B) shows that all three peptides have a similar sequence towards the C-terminus along with the presence of GAS motif, a conserved site for cleavage where Ser residue can be a part of core peptide and could ultimately be amalgamated into the six-membered heterocyclic ring, similar to the Thiomuracin A (Bennallack and Griffiths 2017). DeepRiPP exhibited the cleaved sequence of the first two putative peptides after Ala but for the third one, the cleavage site is after Ser of the GAS motif. The difference in leader cleavage site predictions of putative peptides and that of the known and characterized thiopeptides is the outcome of how the software is programmed and experimental confirmation of exact leader cleavage sites is required.

3.2.2 *Nocardia arthritidis* NBRC 100137: *N. arthritidis* is an aerobic actinomycete that was first isolated from the sputum of an aged Japanese patient suffering from lung nocardiosis and rheumatoid arthritis. It is a

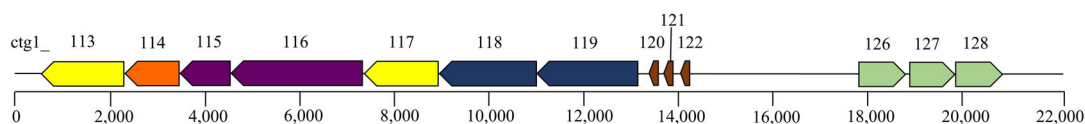
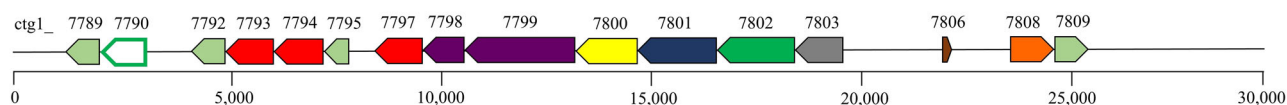
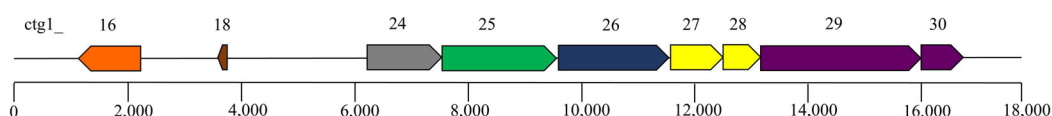
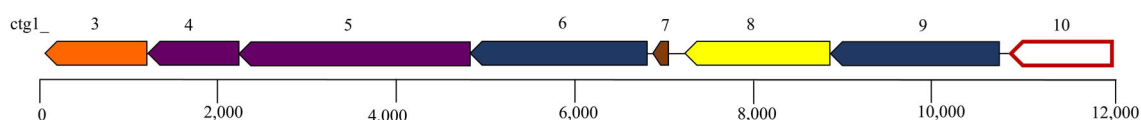
T. onikobensis* NBRC 111776**N. arthritidis* NBRC 100137*****S. afghaniensis* ATCC 23871*****S. pathocidini* NRRL B-24287**

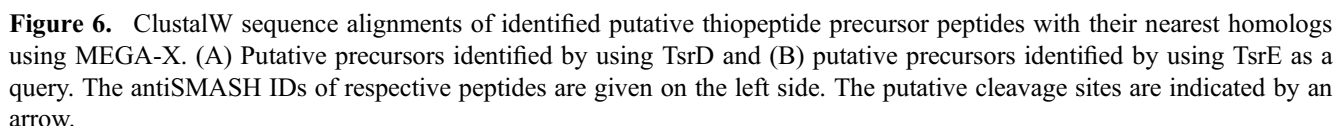
Figure 5. Putative thiopeptide BGCs identified from different bacterial genomes, using TsrE as a query. Gene clusters of the identified BGCs are in comparison to the known thiopeptide, Thiosrepton A (generated by the antiSMASH).

causative agent for potentially life-threatening infection, Nocardiosis (Kageyama *et al.* 2004). The TsrE homolog resulted from PSI-BLAST, had 25% sequence identity; e-value: $2.00\text{E}-16$ with the query (supplementary table 2), and was originally annotated as a ‘hypothetical protein’ (WP_167477808.1) in the NCBI protein database. The antiSMASH data (Cleavage pHMM score: 11.30 and RODEO score: 27) unveiled a BGC (figure 5) containing a single putative precursor peptide (ctg1_7806) and multiple copies of oxygenase and methyltransferase along with core biosynthetic modification enzymes. The putative precursor was similar to Nosiheptide (76.92% sequence identity; e-value: $1\text{E}-11$) in RiPPMiner. The presence of the [4+2] cycloaddition enzyme homolog and putative precursor peptide was observed on the positive strand of the genome sequence by ORFfinder. DeepRiPP exhibited the putative precursor to be a ‘thiopeptide’

with class prediction probability: 100% and showed cleavage site after alanine of the SAS motif.

Pairwise alignment of the putative precursor with close homologs depicts high similarity in the leader region and plenty of conserved Ser, Cys, Thr residues towards the C-terminal (figure 6B). The presence of the SAS motif in alignment evinces that the leader cleavage position will be the same as in Nosiheptide and will make it a new member of thiopeptides.

3.2.3 *Streptomyces afghaniensis* ATCC 23871: *S. afghaniensis* are mesophilic bacteria that build an aerial mycelium. It was isolated from soil in Afghanistan and is a producer of antibiotic taitomycin (Nobuhiko *et al.* 1959). The genomic annotation of *S. afghaniensis* showed the [4+2] cycloaddition enzyme determinant, annotated as ‘hypothetical protein’ (WP_020274901.1) as a TsrE homolog in PSI-BLAST (23.30% identical; e-value:



A highly conserved sequence was observed after pairwise sequence alignment of the putative precursor with Nosiheptide (figure 6B); the main motif for leader

3.2.4 *Streptomyces pathocidini* NRRL B-24287: *Streptomyces pathocidini* is a gram-positive bacteria and is a natural producer of antibiotic pathocidin (Labeda *et al.* 2014). Its genome analysis from the NCBI database (NZ_LIQY01000117.1) unveiled the presence of a TsrE homolog (WP_055471433.1) named as ‘hypothetical protein’ showing 24.33% sequence identity and e-value: 2.00E−05 with TsrE in PSI-BLAST (supplementary table 2). The antiSMASH predicted the BGC (figure 5) with the genes coding for enzyme: split LanB-like dehydratase (*ctgI_4*), [4+2] cycloaddition (*ctgI_3*), two azoline YcaO (*ctgI_6*, 9), azoline dehydrogenase (*ctgI_8*), dehydrogenase (*ctgI_10*) and a putative precursor peptide (*ctgI_7*) (Cleavage pHMM score: 26.10 and RODEO score: 28). The precursor peptide showed 40.38% sequence identity and e-value: 2E−06 with Thiomuracin A in

RiPPMiner. ORFfinder confirmed the presence of TsrE homolog and putative precursor on the negative-strand of the genome sequence. The confirmation for the class of putative peptide was done by DeepRiPP which predicted it as ‘thiopeptide’ with 100% class prediction probability.

ClustalW alignment (figure 6B) of the peptide with Thiomuracin A represents conserved GAS motif and Cys residues towards C-terminal indicating that leader peptide cleavage would occur at the same site as in Thiomuracin A (after Ala of GAS motif). The same cleavage site was predicted by DeepRiPP hence confirming the BGC to be a putative thiopeptide BGC.

4. Discussion

RiPPs are promising alternatives for antibiotics and are effective in fighting against pathogens resistant to several drugs. Thiopeptides are one of the most complex RiPPs due to numerous post-translational modifications and were initially misunderstood as non-ribosomal peptides. Earlier, techniques like product isolation and characterization were used for the discovery of thiopeptides (Morris *et al.* 2009). But in the last decade, bioinformatics tools are being extensively used for the identification of novel RiPPs, as genome mining methodology aids in the unearthing of potentially unique NPs in a culture-free approach, with a probability to cut down the rate for the rediscovery of known metabolites (Baltz 2019). Even the cultures identified through the classical screening and isolation method were later subjected to whole genome sequencing, and bioinformatics tools were used to locate the BGC encoding the isolated product to access its production machinery (Engelhardt *et al.* 2010; Wang *et al.* 2010; Sakai *et al.* 2015). Various strategies to locate the BGC in the sequenced genome of the producer strain have been devised like using the sequence (obtained after PCR amplification) of one of the conserved genes of the BGC (Sakai *et al.* 2015) or using the PCR-amplified gene sequence to probe the cosmid library of the producer strain (Kelly *et al.* 2009; Yu *et al.* 2009; Engelhardt *et al.* 2010; Wang *et al.* 2010). These cumbersome and labour-intensive strategies, ‘Top-down approach’ (Hetrick and van der Donk 2017), are still used the bioinformatics tool to access the flanking region of the probed homologous sequence in thousands of cosmid library clones to define the boundaries of the complete BGC.

The rapid expansion of available genome sequences of a wide variety of bacteria have revealed multiple interesting RiPP clusters, ‘Bottom-up approach’ (Hetrick and van der Donk 2017), that have the potential to encode effective antimicrobials (Cox *et al.* 2015). Since our laboratory has been focussing on the latter ‘Bottom-up approach’ of NPs discovery, many lanthipeptide gene clusters have been identified in our lab using the *in silico* analysis (Singh and Sareen 2014). Our literature survey has revealed that thiopeptides such as lactazole, saalfelduracin, and lactocillin were discovered through genome mining, followed by *in vitro* experiments, to determine the biosynthetic capability and bioactivity of the thiopeptide (Donia *et al.* 2014; Hayashi *et al.* 2014; Schwalen *et al.* 2018).

Actinobacteria is the largest contributing phylum of NPs as it produces nearly three-quarters of microbial products. Predominantly, these bacteria produce a bulk of naturally occurring antibiotics. Typically, actinobacterial genomes are armed with more than one type of RiPPs, of which thiopeptides are equally distributed in both saprophytic and pathogenic strains (Poorinmohammad *et al.* 2019). Even the marine associated actinobacteria have been found to possess RiPPs encoding BGCs (Guerrero-Garzón *et al.* 2020). Recently, gram-positive Bacillales species have also been mined extensively for secondary metabolites and found to be a rich storehouse of novel potential antimicrobial NPs (Zhao and Kuipers 2016). Our genome mining analysis predicted the thiopeptide encoding clusters in Actinobacteria and Ktedonobacteria. Eight RiPP BGCs from different bacterial strains were found, none of which previously is a known producer of thiopeptide. We employed PSI-BLAST to identify putative thiopeptide clusters in different bacteria by taking TsrD and TsrE, from the gene cluster of Thiostrepton A, as a query. By using TsrD as a query, we could identify four RiPP biosynthetic clusters from different strains namely *Planomonospora sphaerica* NRRL 18923, *Micromonospora carbonacea* NRRL 2972, *Thermobifida halotolerans* YIM 90462, *Acrocarpospora macrocephala* NBRC 16266. Two strains, NRRL 18923 and NRRL 2972 were found to encode putative thiopeptide precursors as the peptides in their BGCs were highly similar to well-known thiopeptides and DeepRiPP also predicted their class as Thiopeptide. The other two strains code for several precursor peptides and all the PTM enzymes, but not [4+2] cycloaddition enzyme, which is the class-defining enzyme for thiopeptides (Montalbán-López *et al.* 2020). Although DeepRiPP and RiPPminer predicted their precursors to be thiopeptides, we believe that

these may be post-translationally modified to be LAP, instead of thiopeptide, as most of the enzymes are shared between two BGCs except for the class-defining enzyme. So we put our thoughts together and concluded that using ‘class defining enzymes’ as the driving sequence will be the best approach to dig out homologs otherwise one could be directed towards other RiPP classes BGCs having similar enzymes. Higher surveillance is required to remove the discrepancy in predicting the classes of precursor peptides by DeepRiPP and the identification of whole BGC by the antiSMASH.

The ambiguity in the results led us to change the query sequence from TsrD to TsrE. This query resulted in a long non-redundant list of [4+2] cycloaddition enzyme homologs and from there, four bacteria were randomly selected namely, *Thermogemmatispora onikobensis* NBRC 111776, *Nocardia arthritidis* NBRC 100137, *Streptomyces afghaniensis* ATCC 23871 and *Streptomyces pathocidini* NRRL B-24287. All the bacteria had putative thiopeptide BGCs. The BGC identified in *T. onikobensis* can code for 3 putative precursor peptides and those in *N. arthritidis*, *S. afghaniensis*, and *S. pathocidini* can code for one putative thiopeptide each. All the BGCs contain genes for all enzymes mandatory for core modifications along with other enzymes required for installing side rings and putative precursor peptides that were predicted as thiopeptides by RiPPMiner and DeepRiPP. Therefore, we conclude that only six bacterial strains possess putative thiopeptide gene clusters and thus are capable of forming thiopeptides.

To determine the leader peptide cleavage site and other possible modifications, we aligned the putative precursor peptides with their nearest homologs and did ClustalW alignment using MEGA-X. Upon sequence alignment, we witnessed conserved residues like cysteine, serine, and threonine towards the C-terminus which may attribute high stability and antimicrobial activity to the peptides. In *P. sphaerica* and *M. carbonacea*, the sequence alignment resulted in high similarity with Thiostrepton A which depicts the cleavage sites can be the same, i.e. after methionine and precursor peptide of *P. sphaerica* also showed similarity with Nosiheptide so cleavage for this can be according to Nosiheptide too. However, DeepRiPP displayed some limitations in predicting the leader cleavage site, for the aforementioned putative precursors, as it showed cleavage site for the well-established Thiostrepton A different from the actual. Multiple sequence alignment of precursor peptides from BGC identified from *T. onikobensis* indicates that the

peptides contain GAS motif suggesting that their cleavage and processing can be similar to that of Thiomuracin A. Pairwise alignment of *N. arthritidis* and *S. afghaniensis* with Nosiheptide displayed the conserved SAS motif for cleavage whereas the putative precursor peptide from *S. pathocidini* showed GAS motif when aligned with Thiomuracin A, indicating similar leader cleavage site in both.

A similar methodology as followed earlier in our laboratory for the heterologous expression and characterization of a novel two-component lantibiotic Roseocin (Singh *et al.* 2020) can be adopted for the thiopeptide clusters identified in this study. In conclusion, our results provide confidence in the newly identified BGCs, to be taken up for wet-bench experiments, for the novel thiopeptides discovery. Genome mining coupled with minimally required experimental work can harness the potential of unexplored thiopeptide clusters which can be helpful in the expansion of molecular diversity of thiopeptides with medicinal applications.

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